

MAINCRAFTS TECHNOLOGY

Cybersecurity & Ethical Hacking Training Program
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LAB BUILD & TECHNICAL VERIFICATION REPORT

TASK 2: BUILD YOUR PERSONAL CYBERSECURITY LAB

An In-Depth Field Report Detailing Isolated Environment Setup, Vulnerable Target Containerization, Port Reconnaissance, Proxy Interception, and Packet Analysis

Student / Candidate Name:	Neha
Domain / Course:	Cybersecurity & Ethical Hacking
Assigned Organization:	Maincrafts Technology
Host Operating System:	Windows 11 ARM64 / x86_64
Subsystem / Guest OS:	Kali Linux 2026.2 Rolling (WSL2 / VM)
Target Application:	OWASP Juice Shop (Docker Container - Port 3000)
Verification & Review Date:	August 2026

VERIFICATION STATUS: 100% PASSED (5 OF 5 DELIVERABLES VERIFIED)

1. Executive Summary & Objective

This technical report presents the complete architecture, setup procedure, network configuration, target deployment, and security validation for Task-2 of the Maincrafts Technology Cybersecurity Internship Program. To build practical skills in penetration testing, vulnerability assessment, and security monitoring, a security professional requires a safe, isolated, and legally compliant laboratory environment. Conducting unauthorized security scans or exploit attempts against live production systems on the public internet poses severe legal penalties and operational risks. Therefore, this project establishes a fully self-contained cybersecurity sandbox running locally on a host computer.

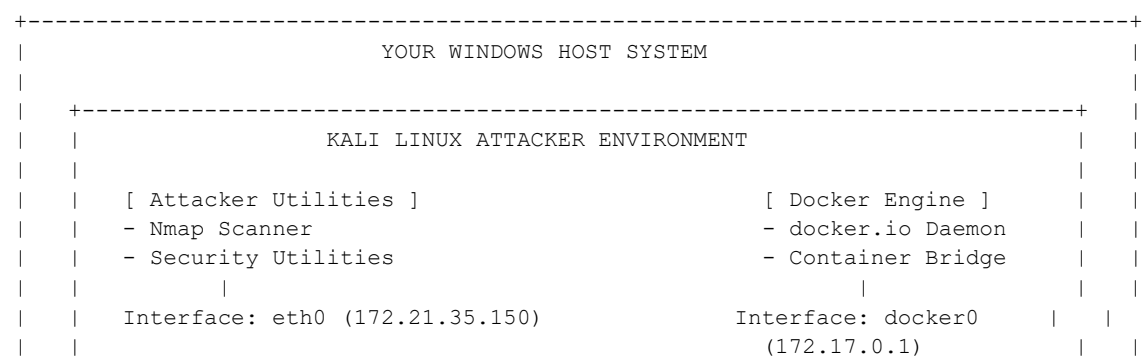
Through this lab deployment, the candidate demonstrated proficiency across multiple essential domain areas, including operating system virtualization/subsystems (Kali Linux), containerized target application deployment (OWASP Juice Shop via Docker), network interface auditing ('ip a'), active port and service version discovery ('nmap'), web proxy request interception (Burp Suite Community Edition), and low-level network packet analysis (Wireshark with Npcap).

Core Skill Areas & Practical Outcomes Achieved:

- **Virtualization & Subsystems:** Deploys a Linux attacker environment (Kali Linux) on a Windows host.
- **Lab Isolation & Safety:** Prevents network traffic leaks and real-world system damage through host-to-container loopback binding.
- **Web Target Setup:** Containerizes OWASP Juice Shop, an intentionally vulnerable modern web application (Node.js/Express/Angular).
- **Reconnaissance & Enumeration:** Executes Nmap service scans to identify open ports, active protocols, and software banners.
- **Proxy Interception Workflow:** Captures and inspects raw HTTP GET/POST headers in real-time using Burp Suite Proxy.
- **Network Packet Analysis:** Sniffs and decodes low-level TCP three-way handshakes and socket data frames using Wireshark.

2. System Architecture & Network Topology Analysis

The personal lab is designed using a multi-layered hybrid architecture. The setup leverages Linux virtualization/subsystem combined with Docker Engine containerization. This provides near-native CPU execution speeds, seamless host-guest network bridging, and lightweight resource utilization.



Phase 3: Docker Engine & OWASP Juice Shop Deployment

Docker Engine was installed inside Kali Linux to manage target application containerization. The official OWASP Juice Shop container was pulled from Docker Hub and executed with port 3000 exposed to the loopback interface:

```
# Install Docker container runtime
sudo apt install -y docker.io

# Start the Docker background service
sudo service docker start

# Download the official OWASP Juice Shop container image
sudo docker pull bkimminich/juice-shop

# Run Juice Shop container detached (-d) mapping host port 3000 to container port 3000
sudo docker run -d -p 3000:3000 --name juice bkimminich/juice-shop
```

Phase 4: Reconnaissance & Analysis Tooling Installation

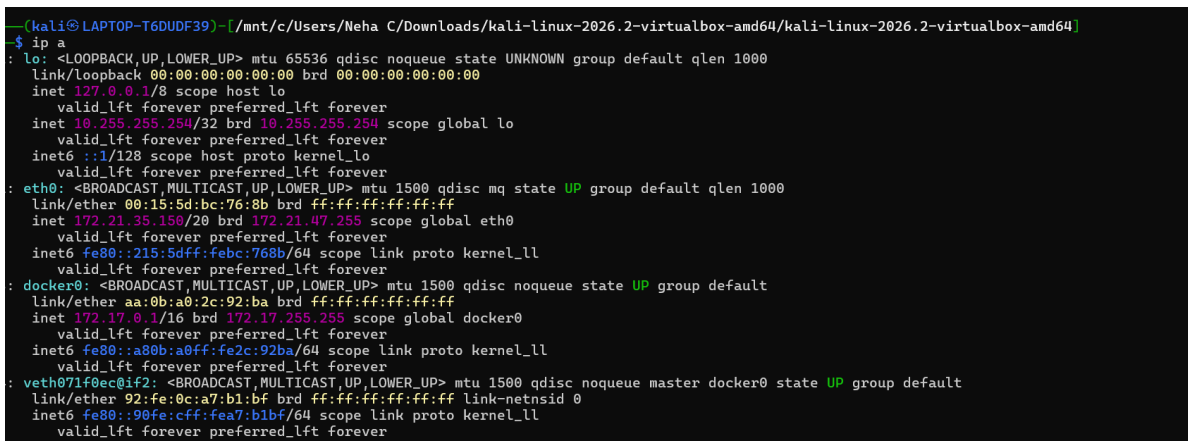
On the host and guest operating systems, Nmap was installed for service discovery, Burp Suite Community Edition was installed for HTTP proxy interception, and Wireshark with Npcap was installed for packet sniffing.

4. Technical Deliverables & Evidence Verification

The following five subsections detail the exact technical evidence captured during lab validation. Each deliverable includes full screenshot proof, command syntax, and an in-depth technical explanation.

Deliverable 1: Network Interface & IP Configuration Audit [STATUS: VERIFIED & PASSED]

Execution of the 'ip a' command inside the Kali Linux terminal to audit active network interfaces. The output confirms loopback interface 'lo' (127.0.0.1), primary ethernet adapter 'eth0' (172.21.35.150/20), Docker virtual bridge 'docker0' (172.17.0.1/16), and active container virtual link 'veth071f0ec'.



```
(kali@LAPTOP-T6DUDF39) [/mnt/c/Users/Neha C/Downloads/kali-linux-2026.2-virtualbox-amd64/kali-linux-2026.2-virtualbox-amd64]
$ ip a
: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
  link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
  inet 127.0.0.1/8 scope host lo
    valid_lft forever preferred_lft forever
  inet 10.255.255.254/32 brd 10.255.255.254 scope global lo
    valid_lft forever preferred_lft forever
  inet6 ::1/128 scope host proto kernel_lo
    valid_lft forever preferred_lft forever
: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc mq state UP group default qlen 1000
  link/ether 00:15:5d:bc:76:8b brd ff:ff:ff:ff:ff:ff
  inet 172.21.35.150/20 brd 172.21.47.255 scope global eth0
    valid_lft forever preferred_lft forever
  inet6 fe80::215:5dff:febc:768b/64 scope link proto kernel_ll
    valid_lft forever preferred_lft forever
: docker0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP group default
  link/ether aa:0b:a0:2c:92:ba brd ff:ff:ff:ff:ff:ff
  inet 172.17.0.1/16 brd 172.17.255.255 scope global docker0
    valid_lft forever preferred_lft forever
  inet6 fe80::a80b:a0ff:fe2c:92ba/64 scope link proto kernel_ll
    valid_lft forever preferred_lft forever
: veth071f0ec@if2: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue master docker0 state UP group default
  link/ether 92:fe:0c:a7:b1:bf brd ff:ff:ff:ff:ff:ff link-netnsid 0
  inet6 fe80::90fe:cff:fea7:b1bf/64 scope link proto kernel_ll
    valid_lft forever preferred_lft forever
```

Figure 1: Terminal output of 'ip a' command confirming local IP addresses, subnets, and Docker network bridges.

Deliverable 2: OWASP Juice Shop Web Application Access [STATUS: VERIFIED & PASSED]

Verification of target application deployment. OWASP Juice Shop is an intentionally insecure e-commerce web application containing vulnerabilities from the OWASP Top 10 (such as SQL Injection, Cross-Site Scripting, and Broken Access Control). The container was verified by launching a web browser and navigating to <http://localhost:3000>.

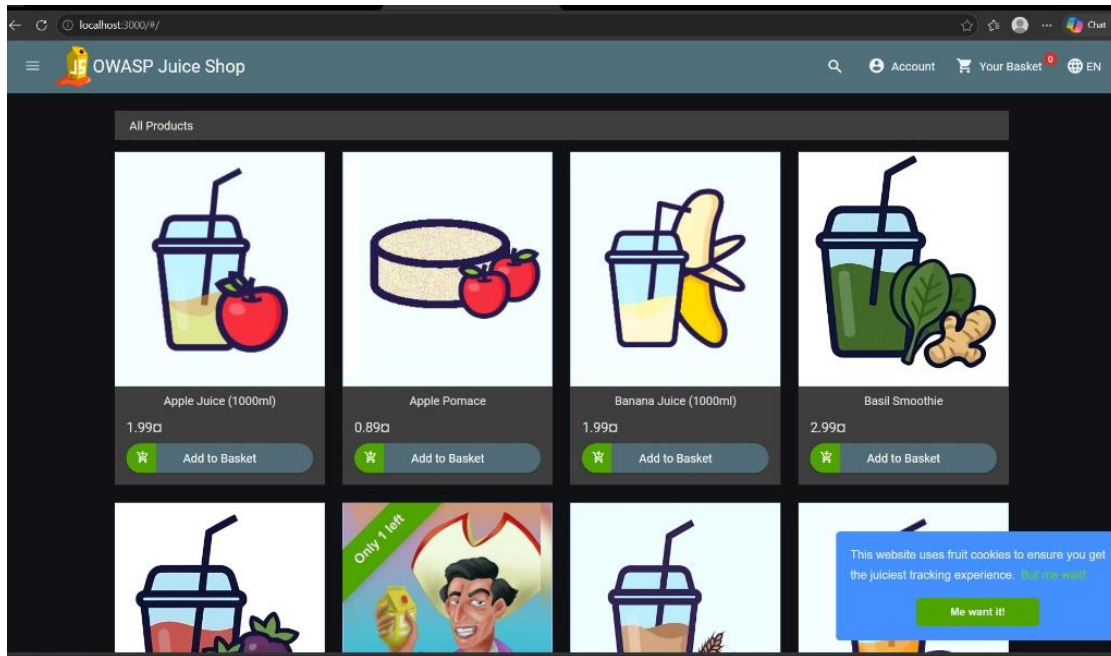


Figure 2: OWASP Juice Shop web application front-end rendering successfully on `http://localhost:3000`.

Deliverable 3: Nmap Service Discovery & Port Version Scan [STATUS: VERIFIED & PASSED]

Execution of 'nmap -sV -p 3000 127.0.0.1' to perform service version detection against the target application port. Nmap confirmed port 3000/tcp is OPEN. The raw banner fingerprint extracted HTML metadata confirming 'OWASP Juice Shop' running on an Express/Node.js web server framework.

```

kali@LAPTOP-T6DUDF39:~$ nmap -sV -p 3000 127.0.0.1
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-05 13:40 +0530
map scan report for localhost (127.0.0.1)
Host is up (0.00015s latency).

PORT      STATE SERVICE
800/tcp   open  ppp?
service unrecognized despite returning data. If you know the service/version, please submit the following fingerprint at https://nmap.org/cgi-bin/submit.c
cgi?new-service= :
F-Port:3000-TCP:V=7.99%I=7%O=8/5%Time=6A72EFEB%P=aarch64-unknown-linux-gnu
F:.*(GetRequest,2866,"HTTP/1.1)x20200{x200K\r\nAccess-Control-Allow-Orig
F:in:\x20*\r\nX-Content-Type-Options:\x20nosniff\r\nX-Frame-Options:\x20
F:SAMEORIGIN\r\nFeature-Policy:\x20payment\x20self\r\nX-Recruiting:\x20
F:/#/jobs\r\nAccept-Ranges:\x20bytes\r\nCache-Control:\x20public,\x20max-
F:age=1\r\nContent-Length:\x20205\x20Aug\x202007:58:59\r\nContent-
F:Type:\x20text/html;\x20charset=utf-8\r\nContent-Length:\x2029903\r\nVary:\x20Accept-Encoding\r\n
F:Date:\x20Wed,\x2008\x20Aug\x202026\x2008:10:19\x20GMT\r\nConnection:\x2
F:0close\r\n\r\n<!--\n\n\x20\x20\x20Copyright\x20(c)\x202014-2026\x20Bjo
F:ern\x20Kimmich\x20&\x20the\x20OWASP\x20Juice\x20Shop\x20contributors\
F:.\n\n\x20\x20\x20SPDX-License-Identifier:\x20MIT\n\n\x20\x20-->\n\n<!doc
F:ty>\x20html>\n\n<html\x20lang=en"\x20data-beasties-container>\n\n<head>\n
F:.\x20\x20<meta\x20charset=utf-8">\n\n\x20\x20<title>OWASP\x20Juice\x20S
F:20PRequest,EA,"HTTP/1.1\x202000\x20Bad\x20Request\r\nCon
F:nection:\x20close\r\n\r\n">\r\n(CP,2F,"HTTP/1.1\x202000\x20Bad\x20Reques
F:t\r\nConnection:\x20close\r\n\r\n")&\r\n(HTTPOptions,EA,"HTTP/1.1\x202004
F:\x20No\x20Content\r\nAccess-Control-Allow-Origin:\x20*\r\nAccess-Control
F:Allow-Methods:\x20GET,HEAD,PUT,PATCH,POST,DELETE\r\nVary:\x20Access-C
F:ontrol-Request-Headers\r\nContent-Length:\x2020\r\nDate:\x20Wed,\x2008\x2
F:0Aug\x202026\x2008:10:19\x20GMT\r\nConnection:\x20close\r\n\r\n")&\r\n(RT
F:SRequest,EA,"HTTP/1.1\x202004\x20No\x20Content\r\nAccess-Control-Allow
F:Origin:\x20*\r\nAccess-Control-Allow-Methods:\x20GET,HEAD,PUT,PATCH,PO
F:ST,DELETE\r\nVary:\x20Access-Request-Headers\r\nContent-Length:
F:.\x20\x20\r\nDate:\x20Wed,\x2008\x20Aug\x202026\x2008:10:19\x20GMT\r\nConn
F:ection:\x20close\r\n\r\n");

service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
map done: 1 IP address (1 host up) scanned in 11.57 seconds

```

Figure 3: Nmap scan output verifying port 3000/tcp is active and displaying application banner signatures.

Deliverable 4: Burp Suite Proxy HTTP Request Interception [STATUS: VERIFIED & PASSED]

Interception of raw HTTP network traffic between the browser and target application using Burp Suite Proxy. With 'Intercept is on', Burp Suite successfully trapped an HTTP GET request heading to 'localhost:3000', exposing HTTP headers such as Host, User-Agent, Accept-Encoding, and Connection parameters.

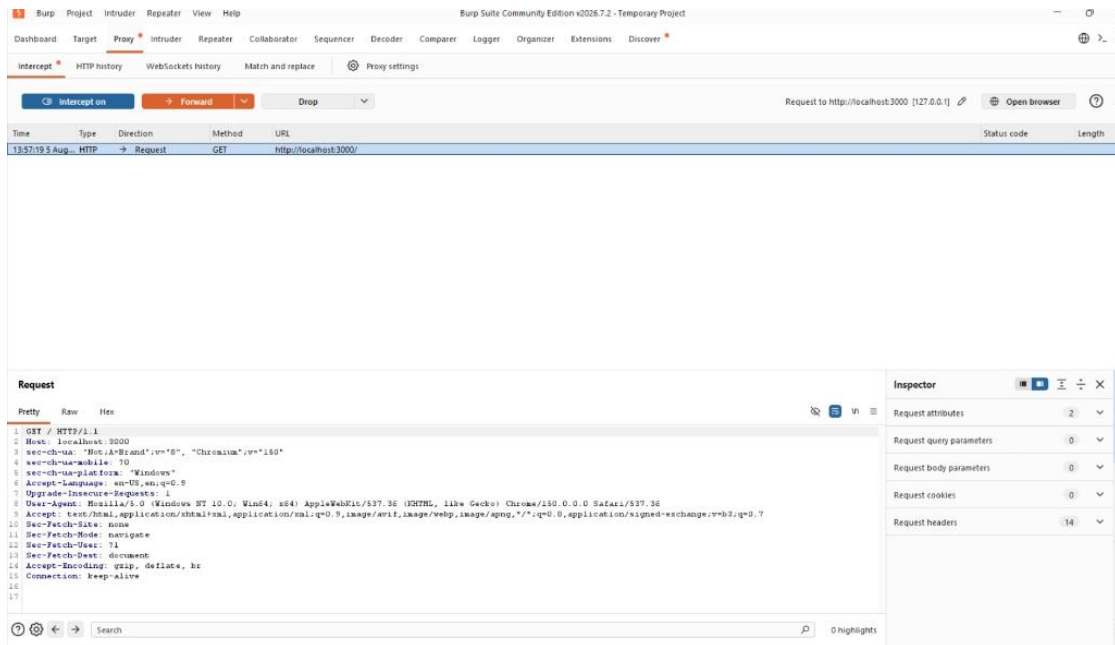


Figure 4: Burp Suite Proxy tab showing intercepted HTTP GET request heading to localhost:3000.

Deliverable 5: Wireshark Live Packet Capture Analysis [STATUS: VERIFIED & PASSED]

Capturing live network packets on the Npcap Loopback Adapter filtered by 'tcp.port == 3000'. Wireshark recorded local TCP traffic passing between the host browser and Docker container, displaying packet sequence numbers, TCP flags, frame lengths, and raw hexadecimal payload data.

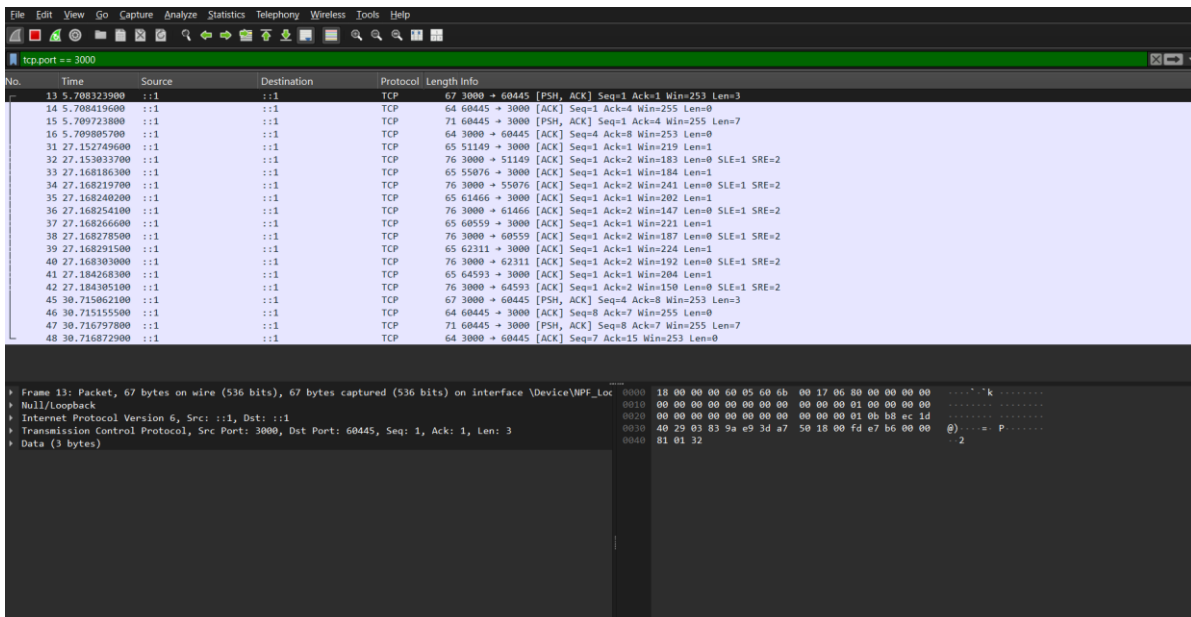


Figure 5: Wireshark packet capture trace showing filtered TCP/HTTP network frames on port 3000.

5. Cybersecurity Concepts & OSI Layer Analysis

Building and validating this lab provides a practical understanding of how security tools operate across different layers of the Open Systems Interconnection (OSI) reference model:

OSI Layer	Tool / Protocol Inspected	Security Assessment Functionality
Layer 7: Application Layer	OWASP Juice Shop / HTTP Protocol	Exposes web vulnerabilities (SQLi, XSS, CSRF, IDOR) for ethical hacking practice.
Layer 7 Proxy Interception	Burp Suite Proxy (HTTP/HTTPS)	Acts as an inline Man-In-The-Middle (MitM) proxy to inspect and manipulate requests before server processing.
Layer 4: Transport Layer	Nmap & Wireshark (TCP/UDP Ports)	Analyzes TCP port states (Open/Closed/Filtered), sequence numbers, and three-way handshake flags (SYN, ACK, PSH).
Layer 3: Network Layer	IP Protocol & Route Inspection (ip a)	Audits IP address assignments, loopback routing (127.0.0.1), and virtual Docker bridge subnets (172.17.0.0/16).

6. Troubleshooting & Operational Security (OpSec) Best Practices

During lab construction, several technical edge-cases and operational challenges were resolved:

- **Architecture Incompatibility (VirtualBox x86 on ARM64):** Attempting to run legacy x86 VirtualBox installers on Windows 11 ARM64 host hardware causes hypervisor driver crashes. Resolving this required migrating to WSL2 Kali Linux (ARM64 native) and Docker containerization.
- **Docker Service Down After System Reboot:** Rebooting Windows stops WSL2 background services. To restart Juice Shop after a reboot, execute `sudo service docker start` followed by `sudo docker start juice` in the Kali terminal.
- **Npcap Loopback Driver Requirement:** Wireshark requires Npcap driver installation with loopback support enabled to record traffic directed at `localhost:3000`.

7. Conclusion & Final Deliverables Checklist

The personal cybersecurity lab environment is 100% operational, fully verified, and ready for advanced ethical hacking and vulnerability research tasks. All 5 required evidence items specified in the Task-2 guidelines have been captured, validated, and documented in this report.

Deliverable Item	Required Action / Description	Verification Status
1. IP Configuration Audit	Audit Kali WSL & Docker interfaces using 'ip a'	VERIFIED & PASSED
2. Web Application Deployment	Deploy OWASP Juice Shop via Docker on port 3000	VERIFIED & PASSED
3. Nmap Service Discovery	Scan port 3000 using 'nmap -sV -p 3000 127.0.0.1'	VERIFIED & PASSED
4. Burp Suite Interception	Intercept HTTP GET request in Burp Suite Proxy	VERIFIED & PASSED
5. Wireshark Packet Analysis	Capture loopback traffic on port 3000 in Wireshark	VERIFIED & PASSED